

GridBalance: Complete Technical Research Dossier

Exhaustive Repository Audit, Data Quality, ML Experiments, Robustness & Production Deployment

UCI Load Forecasting Error Analysis Report

1. Overall Test Performance (Sealed Evaluation)

- **Model:** XGBoost Regressor
- **MAE:** 4421.55 kWh
- **RMSE:** 6654.94 kWh
- **R²:** 0.9944
- **sMAPE:** 1.93%
- **MAE Improvement vs Daily Seasonal Naive:** 78.75%

2. Failure Mode Analysis

A. Performance by Demand Quantile

- **Low Demand (Q1):** MAE = 2529.40 kWh
- **Peak Demand (Q5):** MAE = 5327.04 kWh

Observation: Forecast errors scale proportionally with aggregate load magnitude. Absolute errors peak during maximum load hours (e.g. 18:00 - 20:00).

B. Hourly Error Distribution

- **Lowest Error Hour:** Hour 2 (MAE = 1987.11 kWh)
- **Highest Error Hour:** Hour 22 (MAE = 8891.01 kWh)